MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPES OD3 AND OD3W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Voltage regulator

Outline:
- Type OD3 --- 12-7 (EIA)
- Type OD3W --- See figure 1

Base:
- Type OD3 --- B6-3
- Type OD3W --- See figure 1

Envelope:
- Type OD3 --- ST12
- Type OD3W --- T9

Cathode: Glow discharge

Base connections:

Pin No.	---	1	2	3	5	7	8
Element	---	nc	k	(Note 1)	a	(Note 1)	nc

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ebb	Starting voltage	Operating voltage	Operating current	Alt
Unit:	Vdc	Vdc	Vdc	mAdc	ft
Maximum:	---	---	150 (approx)	40	10,000
Minimum:	---	185	---	5	---
TEST CONDITIONS:	300	---	---	--- (Note 2)	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3347	Ionization voltage (1) (with illumination)	3,4	Rp/Ib = 5 to 40 mAdc	0.65	II	Ez	---	180	Vdc
3337	Voltage drop (1)	-	Rp/Ib = 40 mAdc	0.65	II	Eb(1)	---	162	Vdc
3337	Voltage drop (3)	-	Rp/Ib = 5 mAdc	0.65	II	Eb(3)	145	---	Vdc
3335	Regulation (1)	-	Eb(1) - Eb(3)	0.65	II	Reg(1)	--	5.5	Vdc
3278	Noise and oscillation (oscillation test)	-	Esig = 100 mVac; Rℓ = 500 ohms; Ebb/Ib = 5 to 40 mAdc	0.65	II	---	---	---	---
3278	Noise and oscillation (noise test)	-	Ebb/Ib = 40 mAdc; Rℓ = 500 ohms	0.65	II	ΔEB	---	15.0	mVac
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1041	Shock (Type OD3W)	-	900 G	15	S4	---	---	---	---
---	Shock and vibration-fatigue test end points (Type OD3W):								
3347	Ionization voltage (1)	-		---	---	Ez	---	185	Vdc
3337	Voltage drop (2)	-				Eb(2)	---	163	Vdc
3337	Voltage drop (3)	-				Eb(3)	142	---	Vdc
3335	Regulation (1)	-				Reg(1)	---	5.5	Vdc
3335	Regulation (2)	-				Reg(2)	---	4.5	Vdc
3337	Voltage drop (2)	-	Rp/Ib = 30 mAdc	---	---	Eb(2)	---	160	Vdc
3335	Regulation (2)	-	Eb(2) - Eb(3)	---	---	Reg(2)	---	4.0	Vdc
3305	Leakage current	-	Eb = 50 Vdc	---	---	LIb	---	10.0	μAdc
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1111	Base pin solder depth	5		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group A; Rp/Ib = 30 mAdc	---	---	---	---	---	---
---	Life-test end points (500 hours):								
3347	Ionization voltage (1)	-		---	---	Ez	---	185	Vdc
3337	Voltage drop (2)	-		---	---	Eb(2)	---	163	Vdc
3337	Voltage drop (3)	-		---	---	Eb(3)	142	---	Vdc
3335	Regulation (1)	6		---	---	Reg(1)	---	5.5	Vdc
3335	Regulation (2)	-		---	---	Reg(2)	---	4.5	Vdc
1031	Variable-frequency vibration (Type OD3W)	7	No voltages applied	---	---	---	---	---	---
1031	Vibration fatigue (Type OD3W)	7		---	---	---	---	---	---
3347	Ionization voltage (2) (total darkness)	7,8	Rp/Ib = 5 to 40 mAdc	---	---	Ez	---	225	Vdc

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NOTES:

1. Pins 3 and 7 tied internally.
2. Fixed resistors may be used and Ebb varied to give desired current.
3. This test to be performed at the conclusion of the holding period.
4. Test to be conducted with the tube exposed to light intensity of 5 to 50 foot candles.
5. This test applies for flexible leads as well as for rigid leads.
6. The tube shall have reached the end of its life with regard to regulation (1) when that regulation first exceeds the value of 5.50 volts.
7. (B) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double samplin plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
8. (B) Test to be conducted with the tube under test enclosed in a light tight enclosure composed of a nonconducting material. Measurements to be performed after the speciman is held in the enclosure for 24 hours. The tube shall fire within 20 seconds maximum.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Air Force - 80
DSA - ES

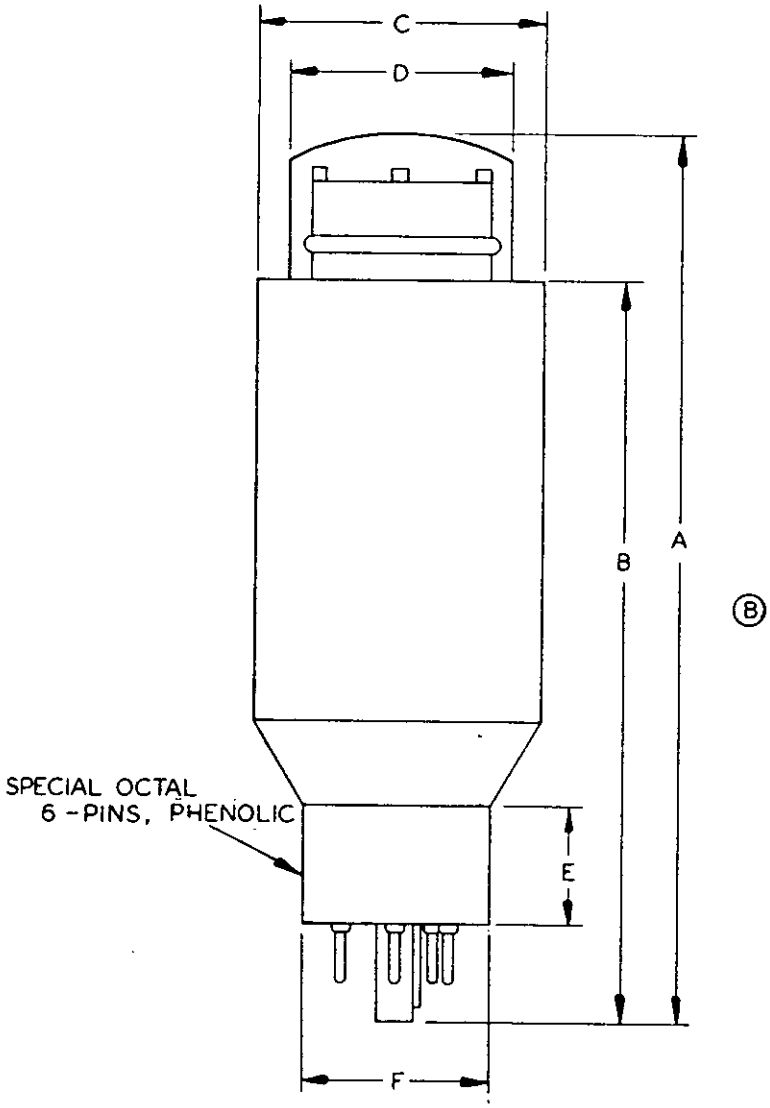
User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3047)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note)	
	Minimum	Maximum
Quality conformance inspection, part 3 (see note 7)		
A	———	4.13 (104.90)
B	———	3.403 (86.44)
C	———	1.57 (39.88)
D	———	1.32 (33.53)
E	.843 (21.41) Nom	
F	1.136 (28.85)	1.176 (29.87)

NOTE:
Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type CD3W.

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